



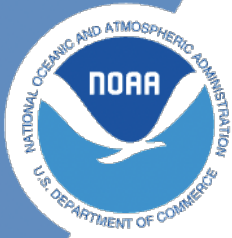
NOAA Technical Memorandum NMFS-XXX-##

NOAA quarto book

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U.S. DEPARTMENT OF COMMERCE
National Oceanic and Atmospheric
Administration
National Marine Fisheries Service
Northwest Fisheries Science Center



**NOAA
FISHERIES**

NOAA quarto book

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1. NOAA Fisheries, Southeast Fisheries Science Center

Table of contents

1	Introduction	1
1.1	Details	1
2	NOAA Fisheries landings data	2
2.1	Data access and upload	2
2.2	Summarize the landings	3
2.3	Plot the summary	5
2.4	Updated Gulf/Atlantic parsing of data	6
2.5	Plots with updated data	9
3	Analysis of international landings estimates from Sea Around Us database	11
3.1	Data access and upload	11
3.2	Read in the high seas landings	11
3.3	Summarize the high seas landings	16
3.4	Read in the landings within exclusive economic zones (EEZs)	19
3.5	Separating the EEZs by area	21
3.6	Summarize catch by regional EEZs	29
3.7	Merge high seas with catch with exclusive economic zones	31
3.8	Plot Western Atlantic catches	34
3.9	Look at Venezuela catches in detail	37
4	SST	47
5	Analysis of ocean temperature trends	48

List of Figures

List of Tables

1 Introduction

Here we will be displaying various data summaries and analyses related to dolphin management in the U.S. South Atlantic.

1.1 Details

The documentation of this data exploration was developed in R, using R Studio and Quarto.

2 NOAA Fisheries landings data

Here we will explore commercial and recreational landings data accessed through the FOSS (Fisheries One Stop Shop) data portal. The advantage of this portal is that commercial and recreational data are available side-by-side in the same units (pounds landed).

2.1 Data access and upload

Data are downloaded from the NOAA Fisheries FOSS site. For the Data Set, we choose both commercial and recreational. For the Years, we select all years from 1980 to the present. For NMFS Region, we select the "NMFS Regions" button and then select the Gulf, South Atlantic, Middle Atlantic and New England from the box. For the Species, we select "Dolphinfish" and "Dolphinfish**." For Report Format, we choose TOTALS BY YEAR/REGION. Hitting the "RUN REPORT" button then generates the data needed for the subsequent analysis, which can be downloaded as a .csv file under the "Actions" button.

We then read in the data file and clean it for analysis (removing the commas from the numbers so that R recognizes the values as numbers). We output the file header to make sure it was read in and that the numbers were converted correctly.

```
# clear workspace
rm(list = ls())

# read in data file
d <- read.table("data/FOSS_landings.csv", header = T, sep = ",", skip = 1)
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94,984	55,676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0

2 NOAA Fisheries landings data

4	1980	South Atlantic	76,042	45,852	Commercial	34
5	1981	Gulf	73,075	46,951	Commercial	33
6	1981	Gulf	1,559,544		Recreational	707

```
# take commas out of file -----
d$Pounds <- as.numeric(gsub(",", "", d$Pounds))
d$Dollars <- as.numeric(gsub(",", "", d$Dollars))
d$Metric.Tons <- as.numeric(gsub(",", "", d$Metric.Tons))
head(d)
```

	Year	Region.Name	Pounds	Dollars	Collection	Metric.Tons
1	1980	Gulf	94984	55676	Commercial	43
2	1980	Middle Atlantic	400	256	Commercial	0
3	1980	New England	200	54	Commercial	0
4	1980	South Atlantic	76042	45852	Commercial	34
5	1981	Gulf	73075	46951	Commercial	33
6	1981	Gulf	1559544	NA	Recreational	707

2.2 Summarize the landings

We then summarize the landings (in pounds) by year, sector, and region. The result is a new data frame with each row representing a year and each column representing the total landings for each region and sector.

```
# compile by year for sectors and regions -----
d$Collection <- factor(d$Collection, levels = c("Recreational", "Commercial"))
tab <- tapply(d$Pounds, list(d$Year, d$Collection, d$Region.Name), sum, na.rm = T)
tab[is.na(tab)] <- 0
#head(tab)

dat <- data.frame(cbind(tab[, , "Gulf"], tab[, , "South Atlantic"],
                        tab[, , "Middle Atlantic"], tab[, , "New England"]))
names(dat) <- c("rec_Gulf", "com_Gulf", "rec_SA", "com_SA", "rec_MA", "com_MA", "rec_NE", "com_NE")
dat
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
1980	0	94984	0	76042	0	400	0	200
1981	1559544	73075	8421898	59037	0	0	0	0

2 NOAA Fisheries landings data

1982	4197503	162457	7573035	143960	2097	200	0	0
1983	2162936	190808	9134022	127986	45927	1400	0	400
1984	1189573	282976	7265430	158901	0	1700	0	400
1985	2542089	269477	6684821	142684	113666	5000	0	4800
1986	5540003	492212	6190187	189747	125653	4200	0	200
1987	4209638	403703	6883546	212905	121433	13400	0	1100
1988	1014312	491610	3134070	212709	130240	26600	2196	17800
1989	12551461	1029628	13273504	426392	1386752	81700	437	15300
1990	13389540	1098819	9283973	514980	624928	69106	2714	14233
1991	13760600	1674124	14756442	650333	722090	90722	6294	9816
1992	12481422	714369	5893700	326831	860112	74423	1411	10315
1993	6873820	596967	7077105	520512	3001335	99420	200927	27069
1994	5448709	679903	10554836	618614	1400635	101044	12586	12328
1995	11142952	1177325	14318412	1164608	1096473	198425	24061	13811
1996	7086006	1047020	9128895	548682	1077364	43581	0	7788
1997	19958088	989648	12284366	877303	1422057	105559	0	13265
1998	9324186	422501	7312090	406368	2920384	86502	120322	11553
1999	7329953	612445	13973364	504263	1005852	98927	4458	5990
2000	5641491	582576	16049049	537077	1433991	36048	0	6545
2001	7355385	368503	14945284	470897	477081	51342	5189	29134
2002	3513226	290601	13794853	436472	1295470	83534	600948	14836
2003	4481937	310836	12112718	459351	929763	46851	129513	21275
2004	2402315	435341	9707678	555973	427145	31916	0	29042
2005	2537132	207695	10299670	376989	437472	19703	0	19982
2006	3290474	224668	12805952	433121	887684	27724	0	19645
2007	3754513	370397	13164377	740255	241212	79832	19542	44707
2008	3937153	384954	10417536	627939	264017	57764	0	16328
2009	3860527	496701	13251102	1007355	1538112	55271	0	92010
2010	1216198	111790	8543542	572031	813738	35869	0	14939
2011	3023914	309324	10774812	543811	2173136	15376	187289	26587
2012	3229090	146256	9983276	704240	755558	44918	26892	46171
2013	5425636	187966	8483569	583556	302421	41936	226988	18564
2014	3074373	201943	11925345	1041856	2129222	58868	183673	61505
2015	6286577	154133	14821772	920454	4501793	36190	1198628	37808
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969
2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421
2022	1664771	15906	6845573	182798	708494	17360	466897	9131

2 NOAA Fisheries landings data

2023 1427518 13305 6630236 125302 734866 28467 122550 8310

2.3 Plot the summary

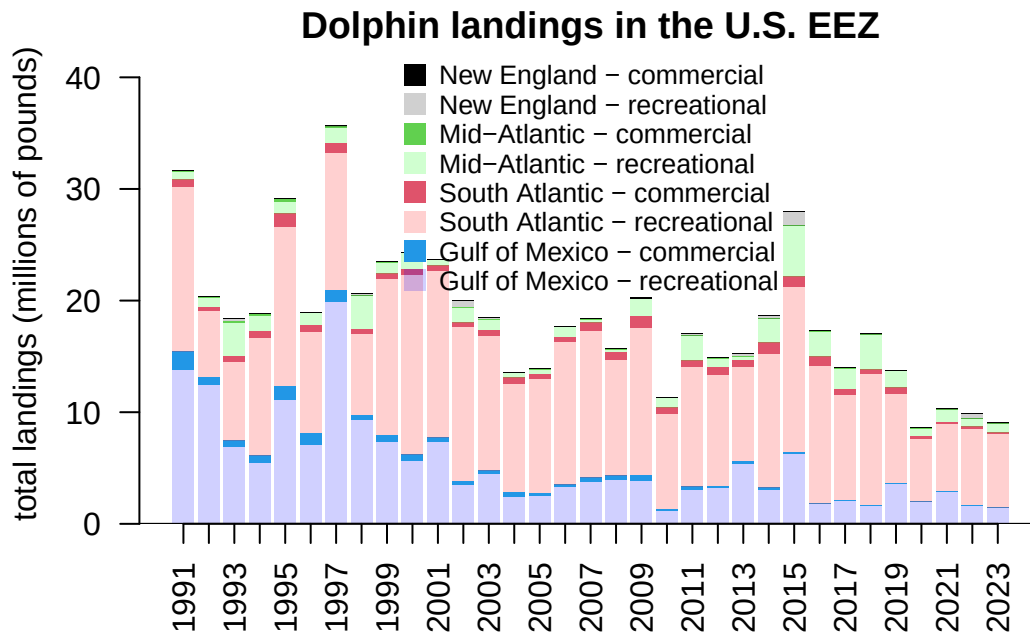
Now we plot the data, using similar color schemes by region. Because the recreational data have high imprecision prior to the 1990s we plot from 1990 to the last year of data.

```
# select years -----
dat <- dat[which(rownames(dat) > 1990), ]

# define colors and plot -----
cols <- c("#0000FF30", 4, "#FF000030", 2, "#00FF0030", 3, "#00000030", 1)

par(mar = c(4, 4, 1, 0), mgp = c(2.5, 1, 0))
yrs <- as.numeric(rownames(dat))

b <- barplot(t(dat)/10^6, names.arg = yrs, col = cols, ylim = c(0, 43),
             border = NA, las = 2, ylab = "total landings (millions of pounds)",
             main = "Dolphin landings in the U.S. EEZ")
legend("top", c("New England - commercial", "New England - recreational",
               "Mid-Atlantic - commercial", "Mid-Atlantic - recreational",
               "South Atlantic - commercial", "South Atlantic - recreational",
               "Gulf of Mexico - commercial", "Gulf of Mexico - recreational"),
       col = cols[8:1], pch = 15, pt.cex = 1.5, bty = "n", y.intersp = 0.9, cex = 0.85)
axis(1, at = b, lab = rep("", length(b)), pos = 0)
abline(h=0)
```



2.4 Updated Gulf/Atlantic parsing of data

There are some issues with the way that the FOSS system splits out Gulf and Atlantic catches by Monroe County, which is on the border between the two jurisdictions. We received data from Mike Larkin (SERO) which has the correct split out of data for Monroe County between the Gulf and Atlantic. We compare the total recreational catch for the Gulf and Atlantic between the FOSS data and the updated data. The numbers are very close; the average difference from the FOSS data is 2%.

```
ml <- read.table("data/RecreationalDolphinLandings_SAandGOM_MLarkin.csv", header = T, sep = ",")
ml <- ml[which(ml$Year > 1990), ]

# select same years for FOSS data set
dat <- dat[which(yrs <= max(ml$Year)), ]

# check that years match
summary(rownames(dat) == ml$Year)
```

Mode TRUE

2 NOAA Fisheries landings data

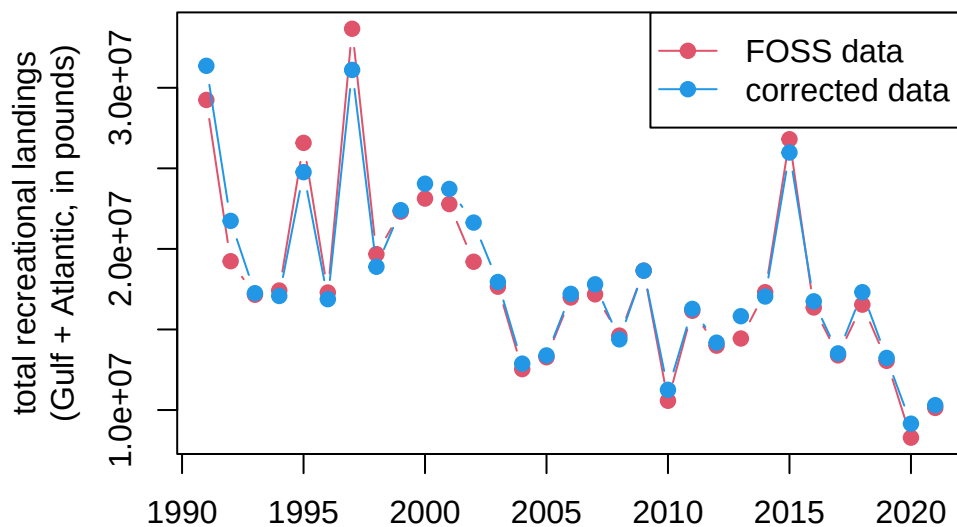
logical 31

```
yrs <- as.numeric(rownames(dat))

# check that totals match
old <- dat$rec_Gulf + dat$rec_SA + dat$rec_MA + dat$rec_NE
new <- ml$GOM + ml$ATL

# plot totals
par(mar = c(4, 6, 2, 1), mgp = c(2.5, 1, 0))
matplot(yrs, cbind(old, new), xlab = "", lty = 1, col = c(2, 4), type = "b", pch = 19,
        ylab = "total recreational landings\n(Gulf + Atlantic, in pounds)",
        main = "Comparison of FOSS data with data corrected for Monroe County")
legend("topright", c("FOSS data", "corrected data"), col = c(2, 4), pch = 19, lty = 1)
```

Comparison of FOSS data with data corrected for Monroe



```
# average % difference between FOSS and updated data
mean((old - new)/old*100)
```

[1] -2.023886

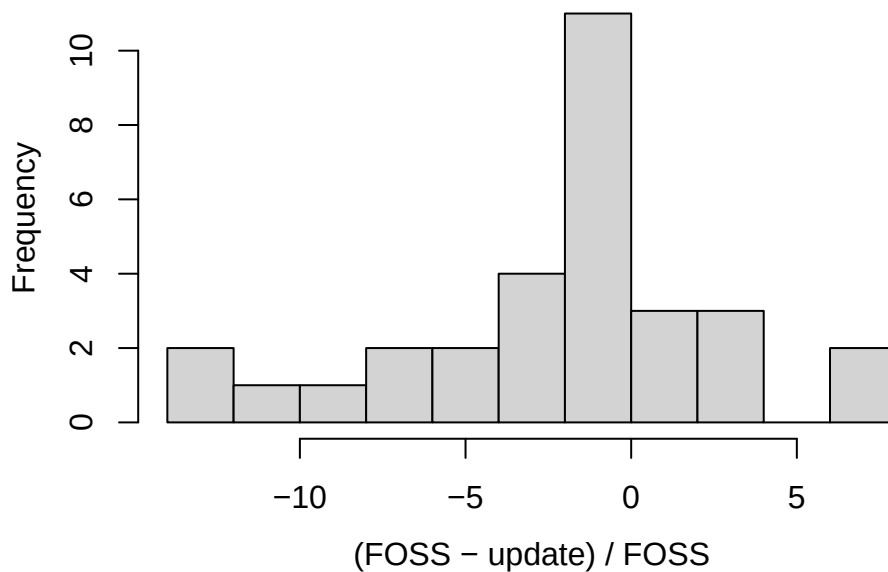
2 NOAA Fisheries landings data

```
# min and max % difference between FOSS and updated data  
range((old - new)/old*100)
```

```
[1] -13.008552  7.587789
```

```
hist((old - new)/old*100, main = "Histogram of percent difference from FOSS data", breaks =  
      xlab = "(FOSS - update) / FOSS")
```

Histogram of percent difference from FOSS data



```
# replace FOSS data with update data  
datnew <- dat  
datnew$rec_Gulf <- ml$GOM  
datnew$rec_SA <- ml$ATL - (datnew$rec_MA + datnew$rec_NE)  
  
# old data set  
tail(dat)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	1777348	37411	12357817	828133	2212513	61336	16499	15747
2017	2064106	65141	9457460	532973	1820541	16179	48607	30969

2 NOAA Fisheries landings data

2018	1626095	56188	11713866	435059	3104018	20514	97107	12667
2019	3567170	83956	7967978	563132	1518629	48224	66	16918
2020	1952241	26332	5653294	260311	679556	14040	0	1880
2021	2893817	26570	6061024	178822	1051785	25318	123038	11421

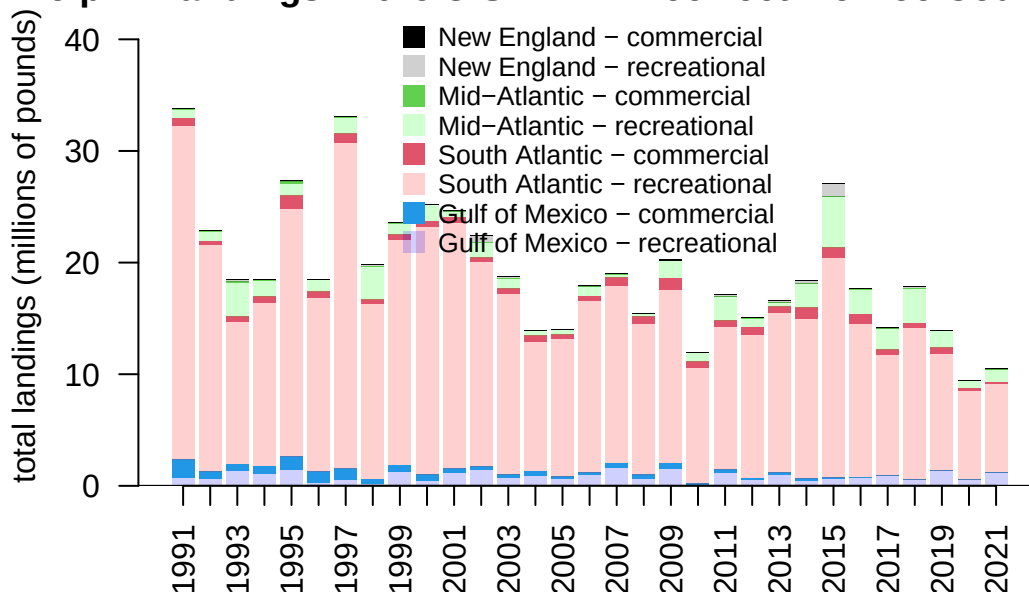
```
# updated data set  
tail(datnew)
```

	rec_Gulf	com_Gulf	rec_SA	com_SA	rec_MA	com_MA	rec_NE	com_NE
2016	751499	37411	13766851	828133	2212513	61336	16499	15747
2017	868489	65141	10777104	532973	1820541	16179	48607	30969
2018	508570	56188	13603875	435059	3104018	20514	97107	12667
2019	1304865	83956	10410599	563132	1518629	48224	66	16918
2020	555695	26332	7916891	260311	679556	14040	0	1880
2021	1174412	26570	7950716	178822	1051785	25318	123038	11421

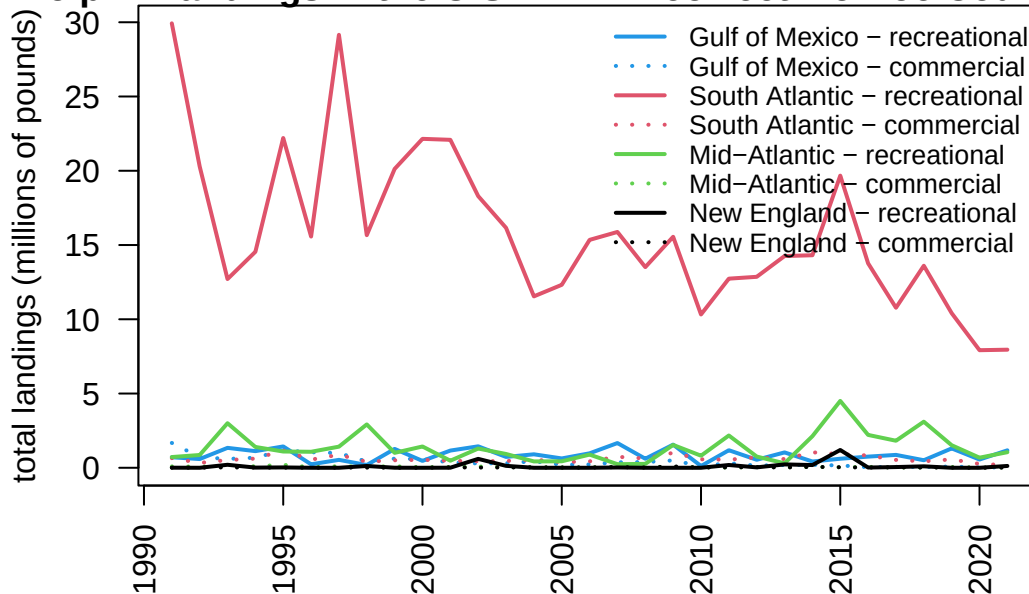
2.5 Plots with updated data

We replace the FOSS values for Gulf and South Atlantic recreational catches with these new values and re-plot the full data set. We can also plot the landings as a line graph so that the relative magnitudes of catch across region and sector are more easily compared.

Dolphin landings in the U.S. EEZ -- correct Monroe Coun



Dolphin landings in the U.S. EEZ -- correct Monroe Coun



3 Analysis of international landings estimates from Sea Around Us database

Here we will explore commercial and recreational landings data as estimated by the Sea Around Us Project. This data set includes reported data for fisheries worldwide as well as reconstructed estimates for unreported landings. For each country, project scientists assemble available data on landings and use additional sociological and fishing data plus expert information and opinion, to produce their best estimates of total landings from all fishing sectors and for all species globally.

3.1 Data access and upload

For catches within each country's exclusive economic zones, data are downloaded from the Sea Around Us site. In the Biodiversity by Taxon Tools and Data, we filter by EEZ for All regions, and then filter by Taxon at the Species level for Common dolphinfish (*Coryphaena hippurus*). After clicking "View graphs" one can access the full data set using the "Download Data" button. The high seas data are accessed in the Basic Search High seas section of the site. The individual regions can be clicked on (e.g., Atlantic, Western Central) and then the data are downloaded using the Download Data button.

The current version of data in use is version 50.1, accessed June 2024 (the data files can be found on github in the SEFSC-dolphin-analyses/data /SAU/ folder). This was still the most recent version of data available as of June 2025.

3.2 Read in the high seas landings

First we read in the high seas landings for the Western Central Atlantic and Northwest Atlantic zones. These are equivalent to FAO areas 21 and 31.

3 Analysis of international landings estimates from Sea Around Us database

```
# clear workspace
rm(list = ls())

install.packages("pals")
library(pals)

# find data files
hslis <- dir("data/SAU/")[grep("HighSeas", dir("data/SAU/"))]

# identify areas 21 and 31
hslis[1]; hslis[3]
```

```
[1] "SAU HighSeas 21 v50-1.csv"
```

```
[1] "SAU HighSeas 31 v50-1.csv"
```

```
#download data
h1 <- read.csv(paste0("data/SAU/", hslis[1]))
h3 <- read.csv(paste0("data/SAU/", hslis[3]))

# confirm correct areas
#unique(h1$area_name)
#unique(h3$area_name)

hs <- rbind(h1, h3)
```

We extract the data for dolphinfish and look at the characteristics of the landings. Most of the landings are in the Western Central Atlantic, and the primary fishing entities are Venezuela and Saint Lucia. The fishing sector is all industrial, and the vast majority of the catch is reported landings, with only a tiny fraction of discards or unreported landings listed in the database. The gear type is primarily long line.

```
# check species name and filter out dolphinfish
unique(hs$common_name)[grep("dolphin", unique(hs$common_name))]
```

```
[1] "Common dolphinfish"
```

3 Analysis of international landings estimates from Sea Around Us database

```
hs <- hs[which(hs$common_name == "Common dolphinfish"), ]  
  
# look at the categories within each column of data  
apply(hs[1:13], 2, table)
```

\$area_name

Atlantic, Northwest Atlantic, Western Central
113 160

\$area_type

high_seas
273

\$year

1985	1986	1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012
1	1	1	1	1	1	1	6	4	1	5	10	5	8	10	12
2013	2014	2015	2016	2017	2018	2019									
24	22	30	33	33	29	34									

\$scientific_name

Coryphaena hippurus
273

\$common_name

Common dolphinfish
273

\$functional_group

Large pelagics (≥ 90 cm)
273

\$commercial_group

Perch-likes

3 Analysis of international landings estimates from Sea Around Us database

273

\$fishing_entity

Barbados	Belize
6	3
Brazil	Canada
19	18
Guyana	Panama
1	1
Saint Lucia	Saint Vincent & the Grenadines
5	28
Spain	Suriname
80	1
Trinidad & Tobago	USA
14	74
Venezuela	
23	

\$fishing_sector

Industrial
273

\$catch_type

Discards Landings
5 268

\$reporting_status

Reported Unreported
253 20

\$gear_type

artisanal fishing gear	bottom trawl	gillnet
7	9	11
hand lines	longline	mixed gear
17	187	1
pole and line	pots or traps	unknown by source

3 Analysis of international landings estimates from Sea Around Us database

19

9

13

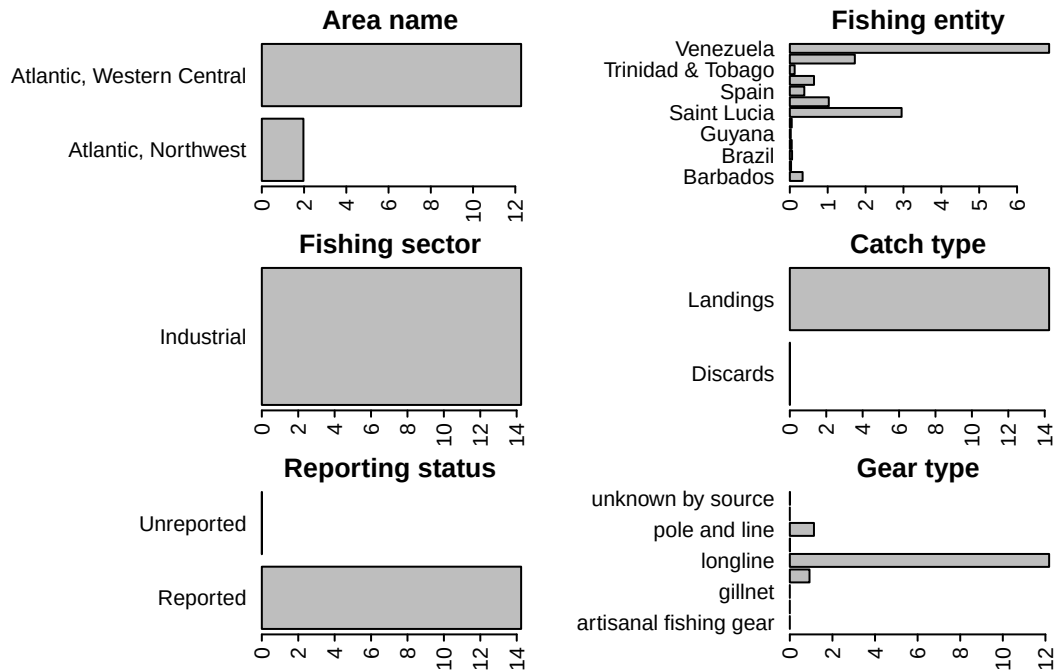
\$end_use_type

	Direct human consumption	Fishmeal and fish oil
5	194	37
Other		
37		

```
# convert from metric tonnes to pounds
hs$pounds <- hs$tonnes * 1000 * 2.20462

# produce barplot of key data fields
par(mar = c(3, 20, 3, 1), mfrow = c(3, 2), mex = 0.5)
barplot(tapply(hs$pounds/10^6, hs$area_name, sum, na.rm = T), las = 2, xlab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$area_type, sum, na.rm = T), las = 2, ylab = "millions of
#barplot(tapply(hs$pounds/10^6, hs$year, sum, na.rm = T), las = 2, ylab = "millions of poun
barplot(tapply(hs$pounds/10^6, hs$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$fishing_sector, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(hs$pounds/10^6, hs$catch_type, sum, na.rm = T), las = 2, xlab = "millions of
barplot(tapply(hs$pounds/10^6, hs$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(hs$pounds/10^6, hs$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```

3 Analysis of international landings estimates from Sea Around Us database



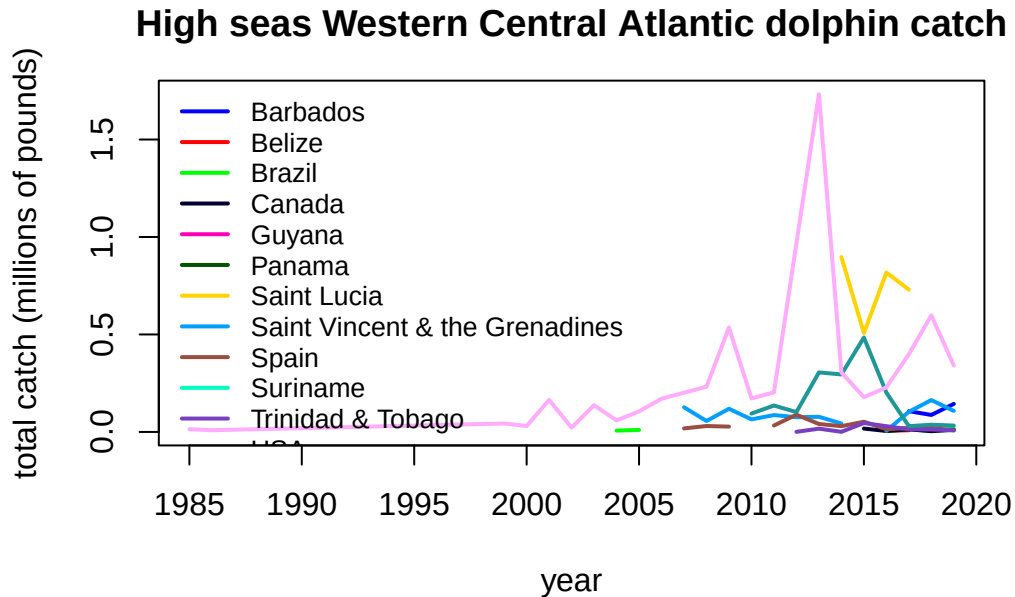
3.3 Summarize the high seas landings

We plot the annual catches by fishing entity to look at trends over time.

```
hs$fishing_entity <- as.factor(hs$fishing_entity)
hs$fishing_entity <- droplevels(hs$fishing_entity)

par(mar = c(5, 5, 3, 1), mfrow = c(1, 1))
tab <- tapply(hs$pounds, list(hs$year, hs$fishing_entity), sum, na.rm = T)
matplot(as.numeric(rownames(tab)), tab/10^6, type = "l", col = glasbey(ncol(tab)), lty = 1,
        main = "High seas Western Central Atlantic dolphin catch")
legend("topleft", colnames(tab), col = glasbey(ncol(tab)), lty = 1, lwd = 2, cex = 0.8, bty
```

3 Analysis of international landings estimates from Sea Around Us database



Finally we output the subset of high seas dolphin catch for the two areas. Because we have already accounted for the USA landings in our national reporting and are aiming to only summarize international landings here, we filter out the USA landings from the Sea Around Us database.

```
hs <- hs[which(hs$fishing_entity != "USA"), ]

hs$year <- factor(hs$year, levels = 1950:2019)
tab <- tapply(hs$pounds, list(hs$year, hs$area_name), sum, na.rm = T)
tab[is.na(tab)] <- 0
tail(tab, 40)
```

	Atlantic, Northwest Atlantic,	Western Central
1980	0.000	0.000
1981	0.000	0.000
1982	0.000	0.000
1983	0.000	0.000
1984	0.000	0.000
1985	0.000	14037.129
1986	0.000	8748.399
1987	0.000	0.000

3 Analysis of international landings estimates from Sea Around Us database

1988	0.000	0.000
1989	0.000	0.000
1990	0.000	0.000
1991	0.000	0.000
1992	0.000	0.000
1993	0.000	0.000
1994	0.000	0.000
1995	0.000	0.000
1996	0.000	0.000
1997	0.000	0.000
1998	0.000	0.000
1999	0.000	43168.296
2000	0.000	30284.627
2001	0.000	164272.223
2002	0.000	22147.477
2003	0.000	137103.181
2004	0.000	66058.258
2005	0.000	115171.863
2006	0.000	170522.101
2007	14687.527	330838.620
2008	28552.636	340863.719
2009	18719.121	662720.661
2010	4510.085	232093.837
2011	21589.424	300498.224
2012	72398.812	1059210.392
2013	34741.604	2471791.396
2014	27922.953	1247462.983
2015	90313.638	710059.125
2016	15319.433	1094893.891
2017	34806.933	1374259.739
2018	29328.358	885814.754
2019	21870.621	632747.157

```
# write summary to merge with EEZ data
write.csv(tab, file = "data/SAU/high_seas_by_year.csv")
```

3.4 Read in the landings within exclusive economic zones (EEZs)

Now we read in the landings reported by each country or jurisdiction's respective EEZ. Because we have extracted dolphin landings for EEZs worldwide, we need to separate out the landings for the Atlantic Ocean basin.

```
rm(list = ls())
d <- read.csv("data/SAU/SAU Taxa 600006 v50-1.csv", header = T, sep = ",") # v. 50.1 - Jun
head(d)
```

	area_name	area_type	year	scientific_name	common_name
1	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
2	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
3	Australia	eez	1950	Coryphaena hippurus	Common dolphinfish
4	Bahamas	eez	1950	Coryphaena hippurus	Common dolphinfish
5	Barbados	eez	1950	Coryphaena hippurus	Common dolphinfish
6	Bermuda (UK)	eez	1950	Coryphaena hippurus	Common dolphinfish
	functional_group	commercial_group	fishing_entity	fishing_sector	
1	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
2	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
3	Large pelagics (>=90 cm)	Perch-likes	Australia	Industrial	
4	Large pelagics (>=90 cm)	Perch-likes	Bahamas	Recreational	
5	Large pelagics (>=90 cm)	Perch-likes	Barbados	Recreational	
6	Large pelagics (>=90 cm)	Perch-likes	Bermuda (UK)	Artisanal	
	catch_type	reporting_status		gear_type	
1	Landings	Reported		longline	
2	Landings	Reported		longline	
3	Landings	Reported		longline	
4	Landings	Unreported	recreational fishing gear		
5	Landings	Unreported	recreational fishing gear		
6	Landings	Reported	small scale lines		
	end_use_type	tonnes	landed_value		
1	Fishmeal and fish oil	0.003373202	3.438170e-05		
2	Direct human consumption	6.739657104	1.040254e+04		
3	Other	0.003373202	3.438170e-05		
4	Direct human consumption	48.665105143	8.262728e+00		
5	Direct human consumption	0.009158559	4.846886e-04		
6	Direct human consumption	1.511085514	3.199403e+03		

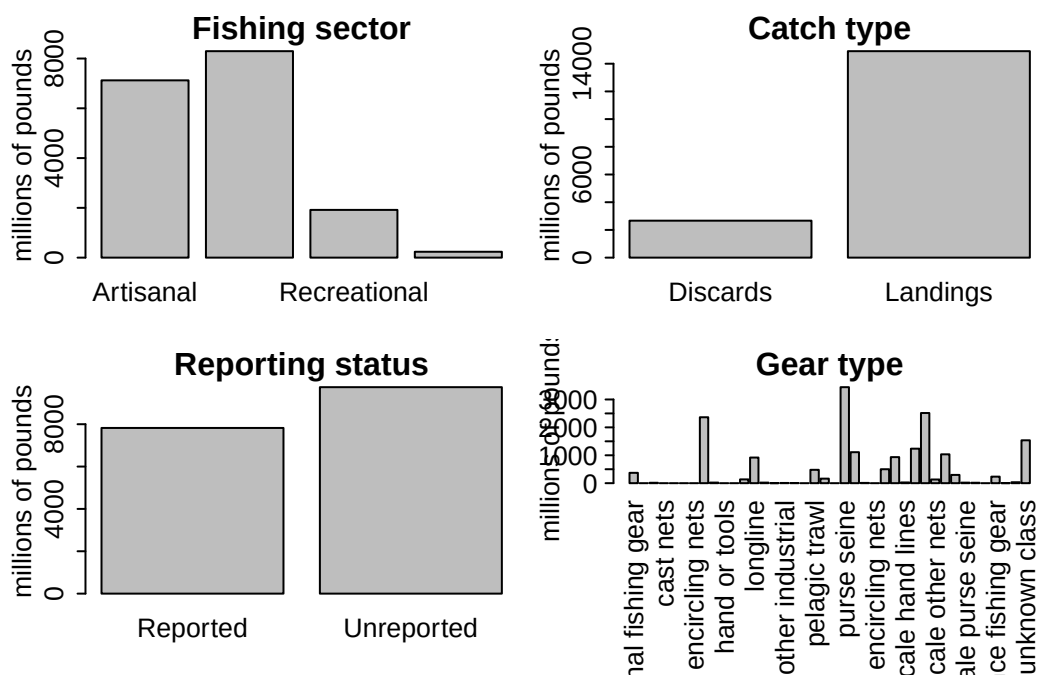
3 Analysis of international landings estimates from Sea Around Us database

```
#dim(d)
#apply(d[1:13], 2, table)

# convert tonnes to pounds
d$lbs <- d$tonnes * 1000 * 2.20462
```

We look at the characteristics of the landings within EEZs. These are largely industrial and artisanal in nature. Landings are the majority of the catch but discards are also present. There are more unreported landings than reported landings and a variety of gear types are used.

```
# look at characteristics of data
par(mfrow = c(2, 2), mar = c(5, 5, 3, 1), mex = 0.5)
barplot(tapply(d$lbs, d$fishing_sector, sum, na.rm = T)/10^6, ylab = "millions of pounds",
barplot(tapply(d$lbs, d$catch_type, sum, na.rm = T)/10^6, ylab = "millions of pounds", main
barplot(tapply(d$lbs, d$reporting_status, sum, na.rm = T)/10^6, ylab = "millions of pounds"
par(mar= c(12, 5, 3, 1))
barplot(tapply(d$lbs, d$gear_type, sum, na.rm = T)/10^6, las = 2, ylab = "millions of pound
```



3.5 Separating the EEZs by area

Now we have to manually assign the different EEZs to respective areas. We separate the geographical areas within the Atlantic Ocean (North and South) and lump all other EEZs into “other oceans.” This latter category is largely the Pacific Ocean and to a lesser extent the Indian Ocean. We remove the high seas catch from this database as it is not ocean-specific and we have extracted the ocean-specific high seas catch previously.

```
# remove the high seas catch from this database
d <- d[which(d$area_type != "high_seas"), ]
d$fishing_entity <- as.factor(d$fishing_entity)
d$fishing_entity <- droplevels(d$fishing_entity)
table(d$area_type)
```

```
eez
63056
```

```
# Western Central Atlantic EEZs
WCA <- c("Bahamas", "Barbados", "Cayman Isl. (UK)", "Dominica", "Dominican Republic", "Grenada",
        "Aruba (Netherlands)", "Haiti", "Guadeloupe (France)", "Jamaica", "Martinique (France)",
        "Montserrat (UK)", "Nicaragua (Caribbean)", "Puerto Rico (USA)", "Saint Kitts & Nevis",
        "Saint Lucia", "Saint Vincent & the Grenadines", "Trinidad & Tobago", "US Virgin Islands",
        "St Barthelemy (France)", "St Martin (France)", "Curaçao (Netherlands)", "Bonaire",
        "Saba and Sint Eustatius (Netherlands)", "Sint Maarten (Netherlands)", "Honduras (Caribbean)",
        "Colombia (Caribbean)", "Mexico (Atlantic)", "Turks & Caicos Isl. (UK)", "Guyana",
        "Antigua & Barbuda", "Belize", "British Virgin Isl. (UK)", "French Guiana", "Cuba",
        "Anguilla (UK)", "Suriname", "Costa Rica (Caribbean)", "Guatemala (Caribbean)",
        "Panama (Caribbean)", "Bermuda (UK)", "Curaçao (Netherlands)")

# Brazil
Bra <- c("Brazil (mainland)", "Fernando de Noronha (Brazil)",
        "St Paul and St. Peter Archipelago (Brazil)")

# Canada
Can <- c("Canada (East Coast)", "Saint Pierre & Miquelon (France)")

# United States
USA <- c("USA (East Coast)", "USA (Gulf of Mexico)")
```

3 Analysis of international landings estimates from Sea Around Us database

```
# Northeast Atlantic EEZs
NEA <- c("Canary Isl. (Spain)", "Spain (mainland; Med and Gulf of Cadiz)", "Spain (Northwest)",
        "Spain (mainland, Med and Gulf of Cadiz)",
        "France (Atlantic Coast)", "Portugal (mainland)", "Azores Isl. (Portugal)")

# Eastern Central Atlantic EEZs
ECA <- c("Madeira Isl. (Portugal)", "Cape Verde", "Equatorial Guinea", "Benin",
        "Sao Tome & Principe ", "Sao Tome & Principe", "Morocco (Central)", "Ghana",
        "Côte d'Ivoire", "Côte d'Ivoire", "Guinea", "Liberia", "Sierra Leone", "Nigeria",
        "Guinea-Bissau", "Senegal", "Cameroon", "Gambia", "Mauritania", "Morocco (South)",
        "Gabon", "Congo; R. of", "Congo, R. of", "Congo (ex-Zaire)")

# Southeastern Atlantic EEZs
SEA <- c("Angola", "Namibia", "South Africa (Atlantic and Cape)",
        "Ascension Isl. (UK)", "Saint Helena (UK)", "Tristan da Cunha Isl. (UK)")

# Southwest Atlantic EEZs
SWA <- c("Trindade & Martim Vaz Isl. (Brazil)", "Uruguay")

# Mediterranean EEZs
Med <- c("Italy (mainland)", "Libya", "Malta", "Syria", "Sicily (Italy)", "Sardinia (Italy)",
        "Balearic Islands (Spain)", "Cyprus (North)", "Cyprus (South)",
        "Gaza Strip", "Greece (without Crete)", "Crete (Greece)", "Lebanon",
        "Turkey (Mediterranean Sea)", "Egypt (Mediterranean)", "Israel (Mediterranean)",
        "Algeria", "Albania", "Croatia", "Montenegro", "Morocco (Mediterranean)",
        "France (Mediterranean)", "Corsica (France)")

# add variable to specify region
d$reg <- "other oceans"
d$reg[which(d$area_name %in% WCA)] <- "Western Central Atlantic EEZs"
d$reg[which(d$area_name %in% Bra)] <- "Brazil EEZ"
d$reg[which(d$area_name %in% Can)] <- "Canadian EEZ"
d$reg[which(d$area_name %in% USA)] <- "United States EEZ"
d$reg[which(d$area_name %in% NEA)] <- "Northeast Atlantic EEZs"
d$reg[which(d$area_name %in% ECA)] <- "Eastern Central Atlantic EEZs"
d$reg[which(d$area_name %in% SEA)] <- "Southeast Atlantic EEZs"
d$reg[which(d$area_name %in% SWA)] <- "Southwest Atlantic EEZs"
d$reg[which(d$area_name %in% Med)] <- "Mediterranean Sea EEZs"
```

Now we will check that we have classified the regions correctly.

3 Analysis of international landings estimates from Sea Around Us database

```
# check region designation
othlis <- unique(d$area_name[which(d$reg == "other oceans")])
print("EEZs of all other oceans: ")
```

```
[1] "EEZs of all other oceans: "
```

```
othlis
```

```
[1] "Australia"
[3] "Taiwan"
[5] "French Polynesia"
[7] "Andaman & Nicobar Isl. (India)"
[9] "Japan (main islands)"
[11] "Korea (South)"
[13] "Nauru"
[15] "Northern Marianas (USA)"
[17] "Marshall Isl."
[19] "Pakistan"
[21] "Philippines"
[23] "Viet Nam"
[25] "Tonga"
[27] "USA (West Coast)"
[29] "Wallis & Futuna Isl. (France)"
[31] "Guatemala (Pacific)"
[33] "Mexico (Pacific)"
[35] "Korea (North, Yellow Sea)"
[37] "Mauritius"
[39] "Christmas Isl. (Australia)"
[41] "Timor Leste"
[43] "Jarvis Isl. (USA)"
[45] "Indonesia (Central)"
[47] "Kiribati (Line Islands)"
[49] "New Caledonia (France)"
[51] "Lord Howe Isl. (Australia)"
[53] "Ecuador (mainland)"
[55] "Fiji"
[57] "Norfolk Isl. (Australia)"
[59] "Pitcairn (UK)"
[61] "Costa Rica (Pacific)"
"China"
"Mayotte (France)"
"Hong Kong (China)"
"Indonesia (Eastern)"
"Japan (Daito Islands)"
"Hawaii Northwest Islands (USA)"
"Niue (New Zealand)"
"Micronesia (Federated States of)"
"Palau"
"Papua New Guinea"
"Seychelles"
"Tokelau (New Zealand)"
"Hawaii Main Islands (USA)"
"Wake Isl. (USA)"
"Samoa"
"Kiribati (Gilbert Islands)"
"Japan (Ogasawara Islands)"
"Korea (North, Sea of Japan)"
"Solomon Isl."
"Johnston Atoll (USA)"
"Palmyra Atoll & Kingman Reef (USA)"
"Howland & Baker Isl. (USA)"
"Indonesia (Indian Ocean)"
"Kiribati (Phoenix Islands)"
"Vanuatu"
"Myanmar"
"El Salvador"
"Malaysia (Peninsula West)"
"Peru"
"Colombia (Pacific)"
"Nicaragua (Pacific)"
```

3 Analysis of international landings estimates from Sea Around Us database

[63] "Panama (Pacific)"	"Thailand (Andaman Sea)"
[65] "Thailand (Gulf of Thailand)"	"Cook Islands"
[67] "Kermadec Isl. (New Zealand)"	"Tuvalu"
[69] "Honduras (Pacific)"	"American Samoa"
[71] "Galapagos Isl. (Ecuador)"	"New Zealand"
[73] "Brunei Darussalam"	"Malaysia (Sabah)"
[75] "Malaysia (Sarawak)"	"Guam (USA)"
[77] "Clipperton Isl. (France)"	"South Africa (Indian Ocean Coast)"
[79] "Singapore"	"Easter Isl. (Chile)"
[81] "Madagascar"	"Réunion (France)"
[83] "Chagos Archipelago (UK)"	"Comoros Isl."
[85] "Mozambique Channel Isl. (France)"	"Kenya"
[87] "Mozambique"	"Somalia"
[89] "Tanzania"	"Glorieuse Islands (France)"
[91] "Yemen (Socotra)"	"Oman"
[93] "St Paul & Amsterdam Isl. (France)"	"Iran (Sea of Oman)"
[95] "Desventuradas Isl. (Chile)"	"India (mainland)"
[97] "Maldives"	"Tromelin Isl. (France)"
[99] "Cambodia"	"Sri Lanka"
[101] "Malaysia (Peninsula East)"	"Bangladesh"

```
print("U.S. EEZs outside the Atlantic Ocean: ")
```

```
[1] "U.S. EEZs outside the Atlantic Ocean: "
```

```
othlis[grep("USA", othlis)]
```

[1] "Hawaii Northwest Islands (USA)"	"Northern Marianas (USA)"
[3] "Hawaii Main Islands (USA)"	"USA (West Coast)"
[5] "Wake Isl. (USA)"	"Johnston Atoll (USA)"
[7] "Palmyra Atoll & Kingman Reef (USA)"	"Jarvis Isl. (USA)"
[9] "Howland & Baker Isl. (USA)"	"Guam (USA)"

```
print("French EEZs outside the Atlantic Ocean: ")
```

```
[1] "French EEZs outside the Atlantic Ocean: "
```

3 Analysis of international landings estimates from Sea Around Us database

```
othlis[grep("France", othlis)]
```

```
[1] "Mayotte (France)"           "Wallis & Futuna Isl. (France)"
[3] "New Caledonia (France)"     "Clipperton Isl. (France)"
[5] "Réunion (France)"          "Mozambique Channel Isl. (France)"
[7] "Glorieuse Islands (France)" "St Paul & Amsterdam Isl. (France)"
[9] "Tromelin Isl. (France)"
```

```
print("Western Central EEZs:")
```

```
[1] "Western Central EEZs:"
```

```
unique(d$area_name[which(d$reg == "Western Central Atlantic EEZs")])
```

```
[1] "Bahamas"
[2] "Barbados"
[3] "Bermuda (UK)"
[4] "Cayman Isl. (UK)"
[5] "Dominica"
[6] "Dominican Republic"
[7] "Grenada"
[8] "Guadeloupe (France)"
[9] "Haiti"
[10] "Jamaica"
[11] "Martinique (France)"
[12] "Montserrat (UK)"
[13] "Puerto Rico (USA)"
[14] "Saint Kitts & Nevis"
[15] "Saint Lucia"
[16] "Saint Vincent & the Grenadines"
[17] "Trinidad & Tobago"
[18] "US Virgin Isl."
[19] "St Barthelemy (France)"
[20] "St Martin (France)"
[21] "Curaçao (Netherlands)"
[22] "Bonaire (Netherlands)"
[23] "Saba and Sint Eustatius (Netherlands)"
[24] "Sint Maarten (Netherlands)"
```

3 Analysis of international landings estimates from Sea Around Us database

```
[25] "Honduras (Caribbean)"
[26] "Colombia (Caribbean)"
[27] "Mexico (Atlantic)"
[28] "Turks & Caicos Isl. (UK)"
[29] "Costa Rica (Caribbean)"
[30] "Panama (Caribbean)"
[31] "Antigua & Barbuda"
[32] "Belize"
[33] "British Virgin Isl. (UK)"
[34] "Cuba"
[35] "French Guiana"
[36] "Guyana"
[37] "Anguilla (UK)"
[38] "Suriname"
[39] "Venezuela"
[40] "Guatemala (Caribbean)"
[41] "Aruba (Netherlands)"
[42] "Nicaragua (Caribbean)"
```

```
print("Brazil EEZs:")
```

```
[1] "Brazil EEZs:"
```

```
unique(d$area_name[which(d$reg == "Brazil EEZ")])
```

```
[1] "Brazil (mainland)"
[2] "Fernando de Noronha (Brazil)"
[3] "St Paul and St. Peter Archipelago (Brazil)"
```

```
print("Canadian EEZs:")
```

```
[1] "Canadian EEZs:"
```

```
unique(d$area_name[which(d$reg == "Canadian EEZ")])
```

```
[1] "Canada (East Coast)"          "Saint Pierre & Miquelon (France)"
```

3 Analysis of international landings estimates from Sea Around Us database

```
print("United States EEZs:")
```

```
[1] "United States EEZs:"
```

```
unique(d$area_name[which(d$reg == "United States EEZ")])
```

```
[1] "USA (East Coast)"      "USA (Gulf of Mexico)"
```

```
print("Northeast Atlantic EEZs:")
```

```
[1] "Northeast Atlantic EEZs:"
```

```
unique(d$area_name[which(d$reg == "Northeast Atlantic EEZs")])
```

```
[1] "Azores Isl. (Portugal)"
[2] "Spain (mainland, Med and Gulf of Cadiz)"
[3] "Canary Isl. (Spain)"
[4] "Portugal (mainland)"
[5] "France (Atlantic Coast)"
[6] "Spain (Northwest)"
```

```
print("Eastern Central Atlantic EEZs:")
```

```
[1] "Eastern Central Atlantic EEZs:"
```

```
unique(d$area_name[which(d$reg == "Eastern Central Atlantic EEZs")])
```

```
[1] "Cape Verde"           "Equatorial Guinea"
[3] "Sao Tome & Principe"  "Madeira Isl. (Portugal)"
[5] "Togo"                 "Ghana"
[7] "Guinea"               "Liberia"
[9] "Guinea-Bissau"        "Senegal"
[11] "Sierra Leone"         "Morocco (Central)"
[13] "Morocco (South)"      "Côte d'Ivoire"
[15] "Gabon"                "Benin"
[17] "Gambia"               "Mauritania"
[19] "Nigeria"              "Cameroon"
[21] "Congo, R. of"         "Congo (ex-Zaire)"
```


3 Analysis of international landings estimates from Sea Around Us database

```
print("Southeast EEZs:")
```

```
[1] "Southeast EEZs:"
```

```
unique(d$area_name[which(d$reg == "Southeast Atlantic EEZs")])
```

```
[1] "Ascension Isl. (UK)"          "South Africa (Atlantic and Cape)"
[3] "Saint Helena (UK)"           "Angola"
[5] "Namibia"                     "Tristan da Cunha Isl. (UK)"
```

```
print("Southwest EEZs:")
```

```
[1] "Southwest EEZs:"
```

```
unique(d$area_name[which(d$reg == "Southwest Atlantic EEZs")])
```

```
[1] "Trindade & Martim Vaz Isl. (Brazil)" "Uruguay"
```

```
print("Mediterranean Sea EEZs:")
```

```
[1] "Mediterranean Sea EEZs:"
```

```
unique(d$area_name[which(d$reg == "Mediterranean Sea EEZs")])
```

```
[1] "Italy (mainland)"          "Libya"
[3] "Malta"                     "Syria"
[5] "Sicily (Italy)"            "Sardinia (Italy)"
[7] "Balearic Islands (Spain)"  "Tunisia"
[9] "Algeria"                   "Morocco (Mediterranean)"
[11] "France (Mediterranean)"    "Corsica (France)"
[13] "Croatia"                   "Lebanon"
[15] "Albania"                   "Greece (without Crete)"
[17] "Crete (Greece)"           "Cyprus (South)"
```

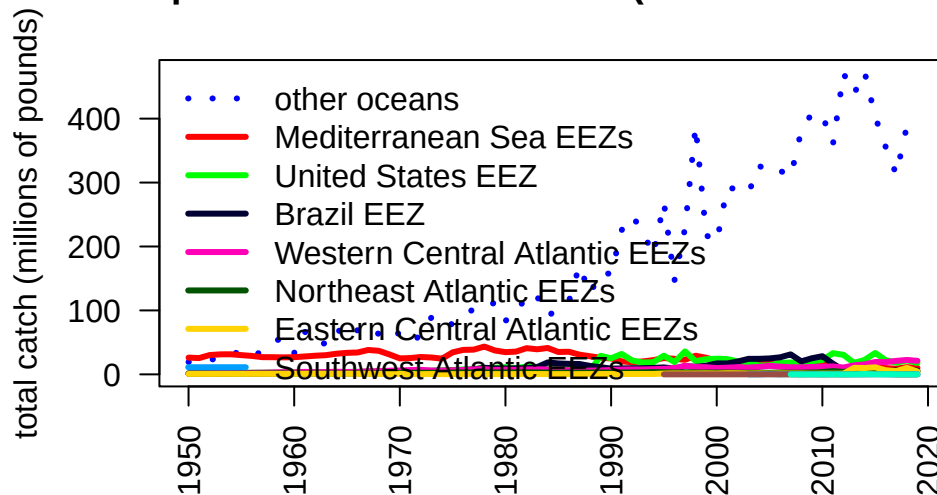
3.6 Summarize catch by regional EEZs

Now we will summarize the catch by region by grouping the catch across the EEZs by region and plot the results. The U.S. landings should be very close to what is reported in Section 2 because the majority of U.S. catch is reported through the various data collection systems that are compiled within Sea Around Us.

```
# summarize EEZ catch by region
tab <- tapply(d$lbs, list(d$year, d$reg), sum, na.rm = T)
#head(tab)
tab2 <- t(tab)
tab3 <- tab2[order(rowSums(tab2, na.rm = T), decreasing = T), ]
tab_glob <- t(tab3)
#head(tab_glob)
yrs <- as.numeric(rownames(tab))

par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_glob/10^6, las = 2, type = "l", lty = c(3, rep(1, 9)), lwd = 3, col = glasbey(ncol(tab_glob)),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Global dolphin catches within EEZs (commercial + recreational)")
legend("topleft", colnames(tab_glob), col = glasbey(ncol(tab_glob)),
      lty = c(3, rep(1, 9)), lwd = 3, bty = "n")
```

Global dolphin catches within EEZs (commercial + recreation)

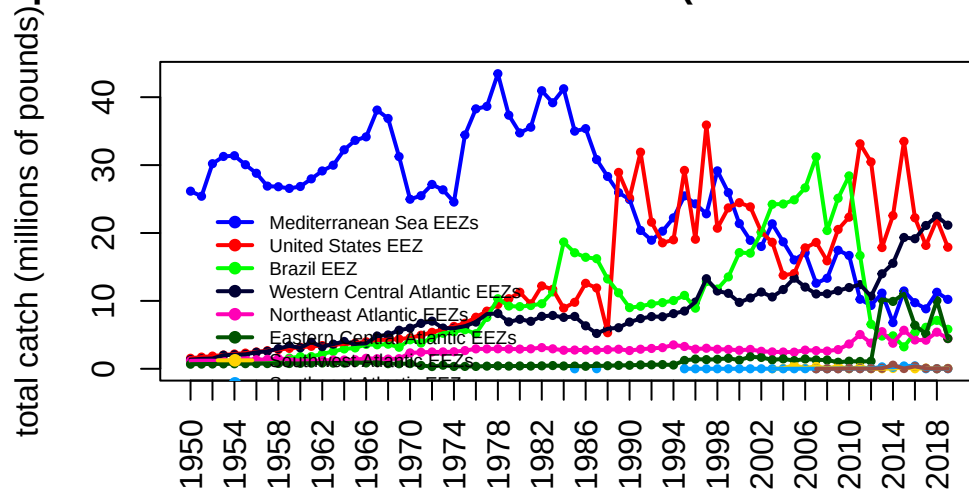


We can see that the EEZ catches globally are dominated by areas outside of the Atlantic ("other oceans") so we will replot the data after moving this category. Here we can see that most of the catches historically were coming from the Mediterranean Sea, though these catches have been declining drastically since the 1970s. Catches in the United States have showed an opposite pattern, with low historical catches that increased in the 1980s (largely due to the introduction of recreational monitoring). Catches in the Western Central EEZs have increased gradually over time.

```
# remove other oceans
#head(tab_glob)
tab <- tab_glob[, -which(colnames(tab_glob) == "other oceans")]
#head(tab)

matplot(yrs, tab/10^6, las = 2, type = "b", pch = 19, cex = 0.5, lty = 1, col = glasbey(ncol(tab)),
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Dolphin catches from the Atlantic EEZs (commercial + recreational)")
matplot(yrs, tab/10^6, las = 2, type = "l", pch = 19, lty = 1, lwd = 2, add = T, col = glasbey(ncol(tab)))
axis(1, at = yrs[seq(1, 90, 2)], las = 2); axis(2); box()
legend(1952, 25, colnames(tab), col = glasbey(ncol(tab)), pch = 19, lty = 1, lwd = 2, bty = "n")
```

olphin catches from the Atlantic EEZs (commercial + recrea



3.7 Merge high seas with catch with exclusive economic zones

Now we will merge in the high seas catch from the steps above into the EEZ catches to get a better picture of total catch in FAO zones 31 and 21. Recall that the U.S. Gulf and Atlantic catches are not necessarily parsed apart correctly in the public databases which get compiled into Sea Around Us, so these will have the same deficiencies as noted in Section 2. Later we will merge in the corrected data.

```
hs <- read.csv("data/SAU/high_seas_by_year.csv")
names(hs)[1] <- "year"
```

```
# check that columns match
#head(tab)
#head(hs)
table(rownames(tab) == hs$year)
```

TRUE

70

3 Analysis of international landings estimates from Sea Around Us database

```
tab[which(is.na(tab))] <- 0
tab_atl <- cbind(tab, hs[2:3])

# check the merge
head(tab_atl)
```

	Mediterranean Sea EEZs	United States EEZ	Brazil EEZ
1950	26162262	1548245	1127242
1951	25410830	1721403	1060734
1952	30206350	1871779	1128992
1953	31275619	2024535	1111008
1954	31383840	2177865	1090492
1955	30056210	2309815	1190424

	Western Central Atlantic EEZs	Northeast Atlantic EEZs
1950	1345027	1173441
1951	1467925	1187386
1952	1510406	1193045
1953	2001465	1141504
1954	2069754	1269523
1955	2130341	1405543

	Eastern Central Atlantic EEZs	Southwest Atlantic EEZs
1950	677552.5	0
1951	693004.4	0
1952	708466.7	0
1953	723939.4	0
1954	739422.5	0
1955	754915.9	0

	Southeast Atlantic EEZs	Canadian EEZ	Atlantic..Northwest
1950	0	0	0
1951	0	0	0
1952	0	0	0
1953	0	0	0
1954	0	0	0
1955	0	0	0

	Atlantic..Western.Central
1950	0
1951	0
1952	0
1953	0
1954	0

3 Analysis of international landings estimates from Sea Around Us database

1955

0

```
# separate all WCA catches
unique(d$reg) # use Western Central, U.S. and Canada

[1] "other oceans"           "Western Central Atlantic EEZs"
[3] "Brazil EEZ"             "Eastern Central Atlantic EEZs"
[5] "Mediterranean Sea EEZs" "Northeast Atlantic EEZs"
[7] "United States EEZ"      "Southeast Atlantic EEZs"
[9] "Southwest Atlantic EEZs" "Canadian EEZ"

dwc <- d[which(d$reg == "Western Central Atlantic EEZs" | d$reg == "United States EEZ" | d$reg == "Canadian EEZ"), ]

#unique(dwc$area_name[which(dwc$reg == "United States EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Canadian EEZ")])
#unique(dwc$area_name[which(dwc$reg == "Western Central Atlantic EEZs")])

# relabel U.S. by area
dwc$reg[which(dwc$area_name == "USA (East Coast)")] <- "United States East Coast"
dwc$reg[which(dwc$area_name == "USA (Gulf of Mexico)")] <- "United States Gulf"
dwc$reg[which(dwc$area_name == "Puerto Rico (USA)")] <- "United States Caribbean"
dwc$reg[which(dwc$area_name == "US Virgin Isl.")] <- "United States Caribbean"
#table(dwc$area_name, dwc$reg)

tab_wc <- tapply(dwc$lbs, list(dwc$year, dwc$reg), sum, na.rm = T)
#head(tab_wc)

# reorder S to N
tab_wc <- tab_wc[, c(5, 2, 4, 3, 1)]
head(tab_wc)
```

	Western Central Atlantic EEZs	United States Caribbean	United States Gulf
1950	1202464	142562.8	309225.3
1951	1299559	168365.6	339921.3
1952	1341943	168463.5	370512.4
1953	1839330	162135.2	400578.5
1954	1901095	168659.3	433479.6
1955	1954511	175830.1	463752.0
	United States East Coast	Canadian EEZ	

3 Analysis of international landings estimates from Sea Around Us database

1950	1239020	NA
1951	1381482	NA
1952	1501267	NA
1953	1623956	NA
1954	1744386	NA
1955	1846063	NA

```
# merge in the high seas catch - check years match
#head(hs)
table(rownames(tab_wc) == hs$year)
```

```
TRUE
70
```

```
high_seas <- hs$Atlantic..Western.Central + hs$Atlantic..Northwest
high_seas[high_seas == 0] <- NA
tab_wc <- cbind(tab_wc, high_seas)
head(tab_wc)
```

	Western Central Atlantic EEZs	United States	Caribbean	United States Gulf
1950	1202464		142562.8	309225.3
1951	1299559		168365.6	339921.3
1952	1341943		168463.5	370512.4
1953	1839330		162135.2	400578.5
1954	1901095		168659.3	433479.6
1955	1954511		175830.1	463752.0

	United States East Coast	Canadian EEZ	high_seas
1950	1239020	NA	NA
1951	1381482	NA	NA
1952	1501267	NA	NA
1953	1623956	NA	NA
1954	1744386	NA	NA
1955	1846063	NA	NA

3.8 Plot Western Atlantic catches

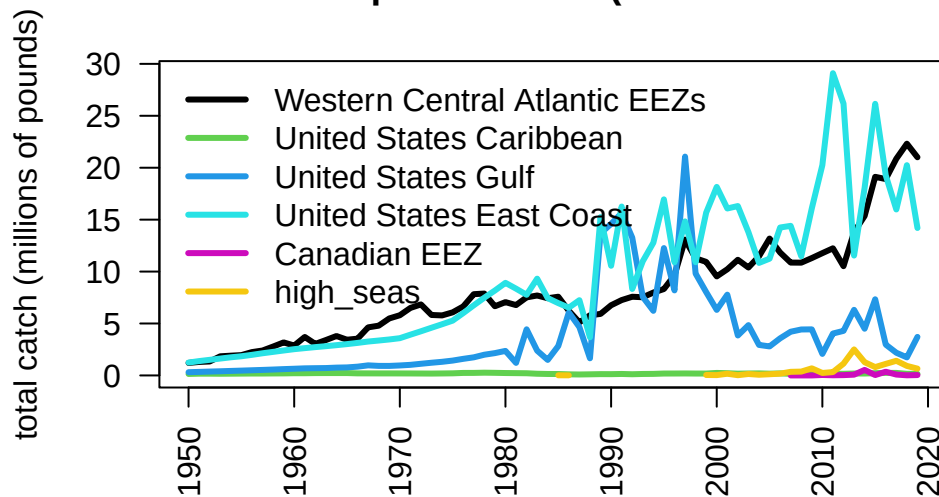
Now we plot the total Western Atlantic catches – both within the EEZs and the high seas – separating by U.S., Canada, and the Western Central Atlantic (i.e., the Wider

3 Analysis of international landings estimates from Sea Around Us database

Caribbean). Most of the catch is coming from the United States and the 40 combined Caribbean jurisdictions which comprise the Western Central Atlantic.

```
par(mar = c(5, 4, 4, 3), mgp = c(3, 1, 0))
matplot(yrs, tab_wc/10^6, las = 2, type = "l", lty = 1, lwd = 3, col = c(1, 3:7),
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches (commercial + recreational)")
legend("topleft", colnames(tab_wc), col = c(1, 3:7), lty = 1, lwd = 3, bty = "n")
```

Western Atlantic dolphin catches (commercial + recreational)



Separating out the mainland United States catches, here are the international and domestic (mainland) catches plotted over time.

```
tab_int <- tab_wc[, c(1, 2, 5, 6)]
head(tab_int)
```

	Western Central Atlantic EEZs	United States Caribbean	Canadian EEZ
1950	1202464	142562.8	NA
1951	1299559	168365.6	NA
1952	1341943	168463.5	NA
1953	1839330	162135.2	NA
1954	1901095	168659.3	NA

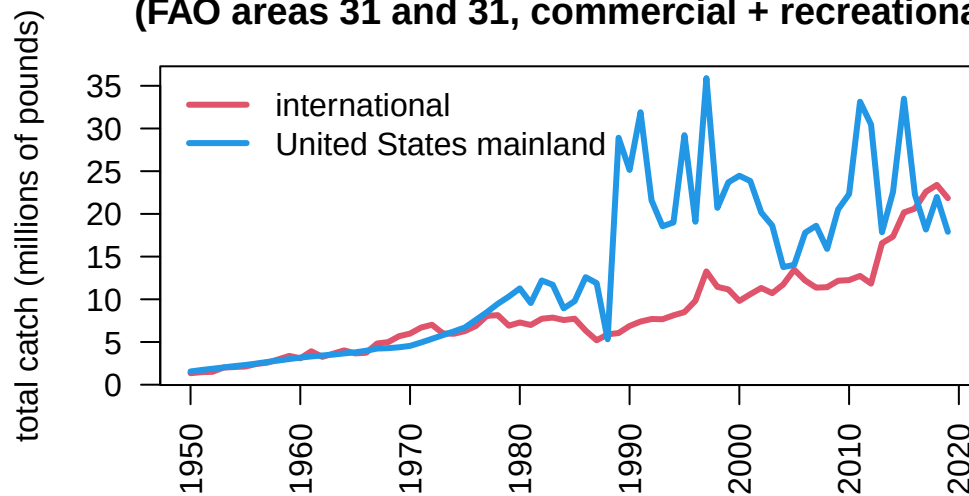
3 Analysis of international landings estimates from Sea Around Us database

1955		1954511	175830.1	NA
	high_seas			
1950	NA			
1951	NA			
1952	NA			
1953	NA			
1954	NA			
1955	NA			

```
int_catch <- round(rowSums(tab_int, na.rm = T), 2)
dom_catch <- rowSums(tab_wc[, c(3, 4)], na.rm = T)

matplot(yrs, cbind(int_catch, dom_catch)/10^6, las = 2, type = "l", lty = 1, lwd = 3, col =
        xlab = "", ylab = "total catch (millions of pounds)",
        main = "Western Atlantic dolphin catches in international vs domestic waters\n(FAO
legend("topleft", c("international", "United States mainland"), col = c(2, 4), lty = 1, lwd
```

Western Atlantic dolphin catches in international vs domestic (FAO areas 31 and 31, commercial + recreational)



```
# output international catches
write.csv(data.frame(int_catch), file = "data/international_catches.csv", row.names = T)
```

3.9 Look at Venezuela catches in detail

Note in the plot above a rapid increase in international catches within the Western Atlantic. This is driven by a sudden increase in catches reported within Sea Around Us within the EEZ. We investigate these catches in detail as it seems odd that a relatively small fleet would be able to ramp up effort so rapidly. Data can be downloaded on a country-specific basis, the catches by taxon in the waters of Venezuela are accessed [here](#).

```
rm(list = ls()) # clear the workspace

# download the data for waters of Venezuela
ven <- read.csv("data/SAU/SAU EEZ 862 v50-1.csv")

# subset dolphinfish from the data
v <- ven[which(ven$common_name == "Common dolphinfish"), ]

apply(v[1:13], 2, table)
```

\$area_name

Venezuela
120

\$area_type

eez
120

\$data_layer

Assigned Tuna RFMO catch	Inferred foreign catch
118	2

\$uncertainty_score
< table of extent 0 >

\$year

1974 1976 1985 1986 1987 1988 1989 1990 1991 1992 1993 1994 1995 1996 1997 1998

3 Analysis of international landings estimates from Sea Around Us database

1	1	1	1	1	1	1	1	1	2	2	2	2	2	2	2	2
1999	2000	2001	2002	2003	2004	2005	2006	2007	2008	2009	2010	2011	2012	2013	2014	
2	2	2	2	2	2	2	2	2	2	2	2	2	3	10	8	
2015	2016	2017	2018	2019												
9	10	11	9	10												

\$scientific_name

Coryphaena hippurus
120

\$common_name

Common dolphinfish
120

\$functional_group

Large pelagics (≥ 90 cm)
120

\$commercial_group

Perch-likes
120

\$fishing_entity

Martinique (France)	Suriname	Trinidad & Tobago
2	1	14
Unknown Fishing Country	Venezuela	
39	64	

\$fishing_sector

Industrial
120

\$catch_type

Landings

3 Analysis of international landings estimates from Sea Around Us database

120

\$reporting_status

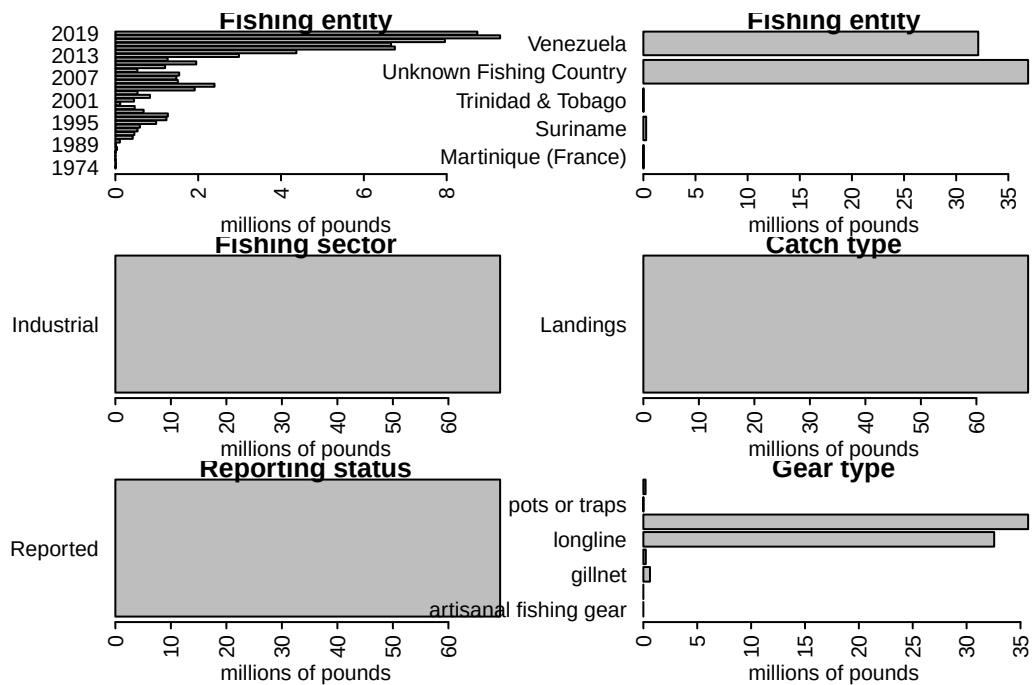
Reported

120

```
#convert to pounds
v$pounds <- v$tonnes * 1000 * 2.20462
#v$fishing_entity <- droplevels(v$fishing_entity)
#v$gear_type <- droplevels(v$gear_type)

# look at the characteristics of the data
par(mar = c(5, 10, 1, 1), mfrow = c(3, 2), mex = 0.5)
#barplot(tapply(v$pounds/10^6, v$area_name, sum, na.rm = T), las = 2, xlab = "millions of p
#barplot(tapply(v$pounds/10^6, v$area_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$year, sum, na.rm = T), las = 2, xlab = "millions of pounds"
barplot(tapply(v$pounds/10^6, v$fishing_entity, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$fishing_sector, sum, na.rm = T), las = 2, xlab = "millions
barplot(tapply(v$pounds/10^6, v$catch_type, sum, na.rm = T), las = 2, xlab = "millions of p
barplot(tapply(v$pounds/10^6, v$reporting_status, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v$pounds/10^6, v$gear_type, sum, na.rm = T), las = 2, xlab = "millions of po
```

3 Analysis of international landings estimates from Sea Around Us database



Note that a large portion of the catch in the database is listed as coming from an “Unknown Fishing Country.” We extract these catches and further investigate the characteristics.

```
# extract the Unknown Fishing Country catch
v1 <- v[which(v$fishing_entity == "Unknown Fishing Country"), ]
apply(v1, 2, table)
```

\$area_name

Venezuela
39

\$area_type

eez
39

\$data_layer

Assigned Tuna RFMO catch

3 Analysis of international landings estimates from Sea Around Us database

39

\$uncertainty_score
< table of extent 0 >

\$year

2013	2014	2015	2016	2017	2018	2019
5	5	5	6	7	5	6

\$scientific_name

Coryphaena hippurus
39

\$common_name

Common dolphinfish
39

\$functional_group

Large pelagics (≥ 90 cm)
39

\$commercial_group

Perch-likes
39

\$fishing_entity

Unknown Fishing Country
39

\$fishing_sector

Industrial
39

\$catch_type

3 Analysis of international landings estimates from Sea Around Us database

Landings

39

\$reporting_status

Reported

39

\$gear_type

artisanal fishing gear	bottom trawl	gillnet
1	5	7
hand lines	longline	pole and line
3	4	7
pots or traps	unknown by source	
5	7	

\$end_use_type

Direct human consumption

39

\$tonnes

1.035978e-01	1.180060e+02	1.181886e+00	1.246067e+03	1.265196e+02	1.359292e-02
1	1	1	1	1	1
1.390137e+00	1.712838e-03	1.847514e+00	2.064341e+01	2.206808e+03	2.257029e+01
1	1	1	1	1	1
2.294627e+03	2.433373e-01	2.437662e-01	2.756940e-02	2.799409e+03	2.974757e-01
1	1	1	1	1	1
3.019718e+01	3.138950e+00	3.187104e+03	3.244301e+01	3.315615e-02	3.447657e+00
1	1	1	1	1	1
3.573195e+03	3.576714e+01	3.856864e-02	3.974389e+01	4.411850e+00	4.991152e+00
1	1	1	1	1	1
5.538519e-03	5.721722e-02	5.721722e-03	6.965107e-02	7.422297e-03	8.655542e-01
1	1	1	1	1	1
8.909156e+02	8.950378e-02	9.087361e+01			
1	1	1			

\$landed_value

3 Analysis of international landings estimates from Sea Around Us database

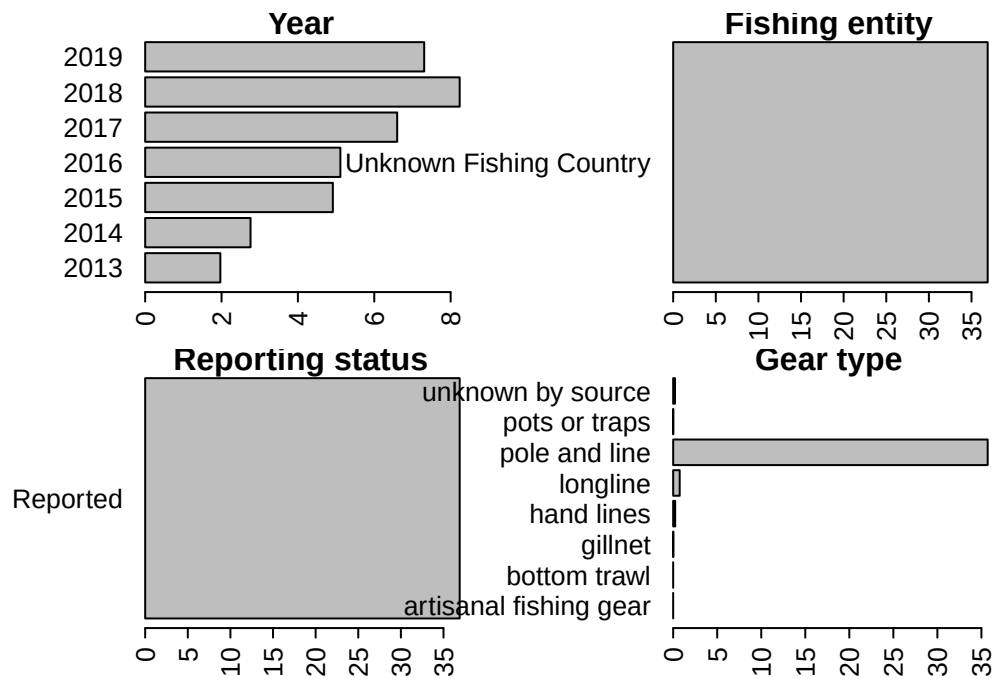
1.061645e+07	1.068040e+04	1.084424e+07	1.103886e+05	1.128144e+03	1.145195e+05
1	1	1	1	1	1
1.166079e+02	1.169389e+04	1.215793e+07	1.276863e+04	1.352301e+05	1.356429e+05
1	1	1	1	1	1
1.602940e+01	1.698258e+04	1.729898e+01	1.729898e+02	2.015819e+02	2.471291e+06
1	1	1	1	1	1
2.517522e+01	2.873676e+02	2.945081e+03	3.045400e+02	3.092009e+05	3.420578e+03
1	1	1	1	1	1
3.606328e+06	4.015198e+05	4.798115e+05	5.124778e+03	5.154963e+01	5.271941e+03
1	1	1	1	1	1
5.809666e+00	6.672035e+06	6.823871e+04	7.001906e+04	7.647414e+01	7.783001e+06
1	1	1	1	1	1
8.253604e+02	8.294236e+02	9.197110e+01			
1	1	1			

\$pounds

1.100359e+04	1.221033e+01	1.261422e+01	1.261422e+02	1.535541e+02	1.636334e+01
1	1	1	1	1	1
1.908218e+03	1.964130e+06	1.973218e+02	2.003418e+05	2.283938e+02	2.601584e+05
1	1	1	1	1	1
2.605610e+03	2.747105e+06	2.789275e+05	2.996721e+01	3.064724e+03	3.776156e+00
1	1	1	1	1	1
4.073067e+03	4.551087e+04	4.865173e+06	4.975890e+04	5.058781e+06	5.364663e+02
1	1	1	1	1	1
5.374118e+02	6.078005e+01	6.171632e+06	6.558209e+02	6.657332e+04	6.920193e+03
1	1	1	1	1	1
7.026353e+06	7.152452e+04	7.309671e+01	7.600773e+03	7.877538e+06	7.885296e+04
1	1	1	1	1	1
8.502920e+01	8.762018e+04	9.726452e+03			
1	1	1			

```
# look at characteristics
par(mar = c(3, 10, 1, 1), mfrow = c(2, 2), mex = 0.6)
barplot(tapply(v1$pounds/10^6, v1$year, sum, na.rm = T), las = 2, xlab = "millions of pound
barplot(tapply(v1$pounds/10^6, v1$fishing_entity, sum, na.rm = T), las = 2, xlab = "million
barplot(tapply(v1$pounds/10^6, v1$reporting_status, sum, na.rm = T), las = 2, xlab = "milli
barplot(tapply(v1$pounds/10^6, v1$gear_type, sum, na.rm = T), las = 2, xlab = "millions of
```


3 Analysis of international landings estimates from Sea Around Us database



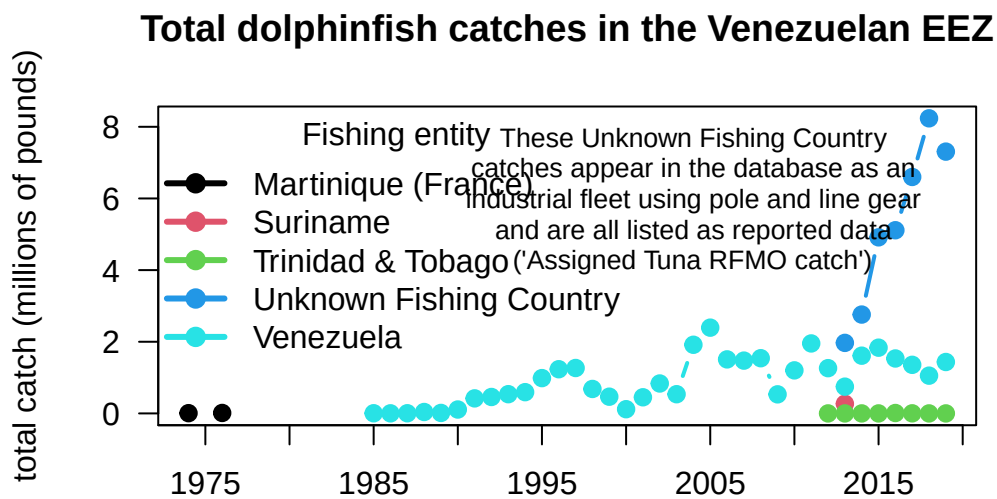
Here is another look at the catches within the venezuelan EEZ, separated by fishing entity.

```
# summarize Venezuela EEZ catches by fishing entity
tab <- tapply(v$pounds, list(v$year, v$fishing_entity), sum, na.rm = T)
head(tab)
```

	Martinique (France)	Suriname	Trinidad & Tobago	Unknown Fishing Country
1974	7320.429	NA	NA	NA
1976	12814.641	NA	NA	NA
1985	NA	NA	NA	NA
1986	NA	NA	NA	NA
1987	NA	NA	NA	NA
1988	NA	NA	NA	NA
Venezuela				
1974	NA			
1976	NA			
1985	4699.563			
1986	2229.484			
1987	4585.610			
1988	39960.149			

3 Analysis of international landings estimates from Sea Around Us database

```
matplot(rownames(tab), tab/10^6, las = 2, type = "b", lty = 1, lwd = 2, pch = 19, col = 1:5,
        xlab = "", ylab = "total catch (millions of pounds)", axes = F,
        main = "Total dolphinfish catches in the Venezuelan EEZ")
axis(1, at = seq(1950, 2020, 5)); axis(2, las = 2); box()
legend("topleft", colnames(tab), col = 1:5, pch = 19, lty = 1, lwd = 3, bty = "n", title =
text(2004, 6, cex = 0.8,
     "These Unknown Fishing Country\ncatches appear in the database as an\nindustrial fleet
```



The very substantial increase in international Western Atlantic dolphin catches can be attributed solely to this Unknown Fishing Country catches which appear in the database beginning in 2013. The documentation of the catch reconstruction can be found in the Sea Around Us documentation files (see Mendoza et al. 2015). We spoke to Jeremy Mendoza and other experts who were involved in the reconstruction of domestic catches in Venezuela, and they noted that the "Assigned tuna RFMO catch" actually emerge from spatially reshuffled landing reports to ICCAT, according to the methods of Coulter et al. (2020). In this work, the Sea Around Us team developed a global spatial reallocation of landings of tunas and associated species that were reported to ICCAT. In short, this work uses the fraction of reported spatially disaggregated data to reallocate all data reported to ICCAT. This step is necessary because there are not many countries that report spatial data to ICCAT in the region, and this method may

3 Analysis of international landings estimates from Sea Around Us database

lead to a significant reallocation of landings from the Caribbean to the Venezuelan EEZ, where the local pole and line fishery is significant (J. Mendoza, personal communication). Thus, it is likely that landings from other EEZs, that were included in the reconstructed catch data layer, could have been reassigned to the Venezuelan EEZ.

Because there may be some overlap between the reconstructed data used in the national EEZ estimates, and the ICCAT reporting that was reassigned, there could potentially be a “double counting” of these catch estimates in the Sea Around Us database. These Unknown Fishing Country catches that lead to a large increase in total estimated catches within the Venezuelan EEZ should be treated with caution at this time, until we can verify with Sea Around Us database managers that the spatial disaggregation method used has not led to an overestimate.

4 SST

5 Analysis of ocean temperature trends

Coming soon